CMOS VLSI Design Project

Design and Analysis of an 8-bit Ripple Carry Adder

Using Cadence Virtuoso

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1 Introduction

An 8-bit ripple carry adder (RCA) is a combinational arithmetic circuit used to add two 8-bit binary numbers. It accepts two operands, A and B, together with an initial carry input, C_{in}, and produces an 8-bit sum and a final carry output, C_{out}.

The circuit is implemented by connecting eight 1-bit full-adder stages in cascade. The carry output of each stage is connected to the carry input of the next higher-order stage. Therefore, a carry generated at a lower bit position can propagate through the subsequent stages.

In this project, an 8-bit ripple carry adder was designed and analyzed using the Cadence Virtuoso design environment. The design flow includes schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction, and pre-layout and post-layout transient simulations. The simulation results are used to examine the output waveforms and evaluate average power consumption and worst-case propagation delay.

2 Objectives

The main objectives of this project are:

- To design an 8-bit ripple carry adder using Cadence Virtuoso.
- To implement the adder using eight cascaded 1-bit full-adder stages.
- To create the schematic and generate a reusable symbol for the complete adder.
- To develop the physical layout of the complete circuit.
- To verify the layout using DRC and LVS.
- To perform RC parasitic extraction using the RCX flow.
- To perform pre-layout and post-layout transient simulations.
- To verify the sum and carry outputs for the applied input combinations.
- To calculate and compare average power consumption and worst-case propagation delay.

3 Theory of an 8-bit Ripple Carry Adder

A ripple carry adder is constructed by cascading full-adder cells. Each full-adder stage adds one bit from operand A, one bit from operand B, and the carry received from the previous stage.

Let the two 8-bit operands be

$$A = (A_7, A_6, \ldots, A_1, A_0), \tag{1}$$

$$B = (B_7, B_6, \ldots, B_1, B_0). \tag{2}$$

The initial carry is $C_0 = C_{in}$ and the final carry is $C_8 = C_{out}$. The sum output is $S = (S_7, S_6, \ldots, S_1, S_0)$.

3.1 Boolean Expressions of Each Full-Adder Stage

For the i-th full-adder stage, where $i = 0, 1, \ldots, 7$, the propagate and generate terms are

$$P_i = A_i \oplus B_i, \tag{3}$$

$$G_i = A_i B_i. \tag{4}$$

The sum output of the stage is

$$S_i = P_i \oplus C_i = A_i \oplus B_i \oplus C_i. \tag{5}$$

The carry output is

$$C_{i+1} = G_i + P_i C_i = A_i B_i + C_i(A_i \oplus B_i). \tag{6}$$

Here, C_i is the carry input of stage i, and C_{i+1} is its carry output.

3.2 Carry Propagation

The carry connection between the eight stages is

$$\begin{aligned} &C_0 = C_{in}, \\ &C_1 \rightarrow \text{carry input of stage 1}, \\ &C_2 \rightarrow \text{carry input of stage 2}, \\ &\quad \vdots \\ &C_7 \rightarrow \text{carry input of stage 7}, \\ &C_8 = C_{out}. \end{aligned}$$

The stages are connected in the order

$$FA_0 \longrightarrow FA_1 \longrightarrow FA_2 \longrightarrow \cdots \longrightarrow FA_7. \tag{7}$$

The carry output of one stage becomes the carry input of the next stage. Since the carry may need to propagate through successive stages, the carry-chain path is an important factor in the worst-case propagation delay.

3.3 Full-Adder Truth Table

The following truth table applies to each individual bit stage.

Table 1: Truth table of an individual full-adder stage

A_i	B_i	C_i	S_i	C_{i+1}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The complete 8-bit adder performs this operation at all eight bit positions. Its numerical result is represented by

$$\{C_{out}, S_7, S_6, \ldots, S_1, S_0\} = A + B + C_{in}. \tag{8}$$

4 Logic-Level Implementation

The 8-bit ripple carry adder is implemented using eight 1-bit full-adder stages. The least-significant stage receives A_0, B_0, and the external carry input C_{in}. Each subsequent stage receives the corresponding operand bits and the carry generated by the preceding stage.

Table 2: Stage connections of the 8-bit ripple carry adder

Stage	Operand inputs	Carry input	Sum output	Carry output
FA_0	A_0, B_0	$C_0 = C_{in}$	S_0	C_1
FA_1	A_1, B_1	C_1	S_1	C_2
FA_2	A_2, B_2	C_2	S_2	C_3
FA_3	A_3, B_3	C_3	S_3	C_4
FA_4	A_4, B_4	C_4	S_4	C_5
FA_5	A_5, B_5	C_5	S_5	C_6
FA_6	A_6, B_6	C_6	S_6	C_7
FA_7	A_7, B_7	C_7	S_7	$C_8 = C_{out}$

The main blocks and signals used in the design are summarized in Table 3.

Table 3: Main blocks and signals used in the ripple carry adder

Block or signal	Quantity	Purpose
1-bit full-adder stage	8	Performs one-bit addition and generates a carry.
Operand A	1 bus	Provides the eight bits A_7 to A_0.
Operand B	1 bus	Provides the eight bits B_7 to B_0.
Carry input	1	Supplies the initial carry C_{in}.
Sum output	1 bus	Provides the eight sum bits S_7 to S_0.
Carry output	1	Provides the final carry C_{out}.

5 Circuit Design

5.1 Schematic Design

The 8-bit ripple carry adder schematic was created using the Cadence Virtuoso Schematic Editor. The circuit accepts two 8-bit operands, $A[7:0]$ and $B[7:0]$, and one carry input, C_{in}. It produces an 8-bit sum, $S[7:0]$, and one carry output, C_{out}.

The eight full-adder stages are connected in cascade. The carry output of each lower-order stage is connected to the carry input of the next higher-order stage. The complete schematic is shown in Figure 1.

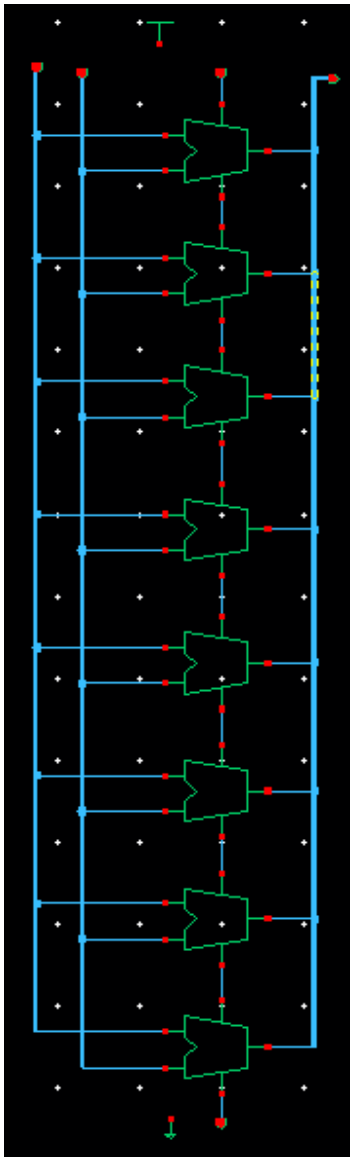


Figure 1: Schematic of the 8-bit ripple carry adder implemented in Cadence Virtuoso.

The implemented circuit follows the stage equations

$$S_i = A_i \oplus B_i \oplus C_i, \tag{9}$$

$$C_{i+1} = A_i B_i + C_i(A_i \oplus B_i), \qquad i = 0, 1, \ldots, 7. \tag{10}$$

5.2 8-bit Ripple Carry Adder Symbol

After completing the schematic, a symbol was generated for the 8-bit ripple carry adder. The symbol provides the operand inputs, initial carry input, sum outputs, and final carry output. The generated symbol is shown in Figure 2.

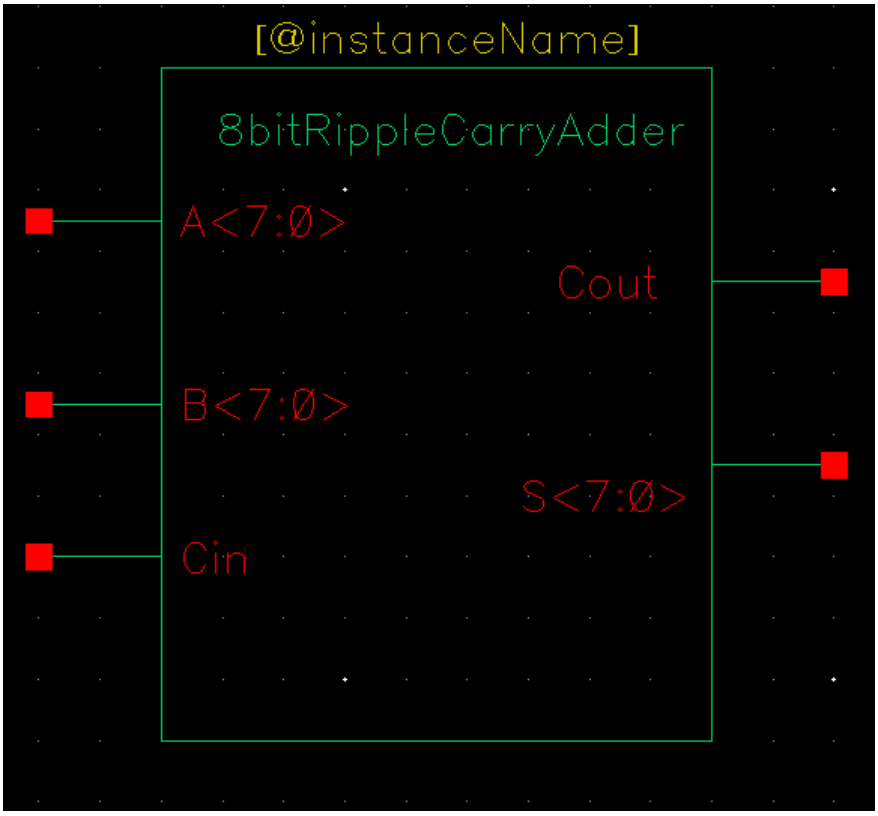


Figure 2: Generated symbol of the 8-bit ripple carry adder.

The symbol can be used as a reusable hierarchical block in a larger digital circuit. The pin names and bus notation should match those used in the Cadence schematic.

6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the 8-bit ripple carry adder was developed using the Cadence Virtuoso Layout Editor. The layout represents the physical placement and interconnection of the devices used in the circuit. It includes the required active or diffusion regions, polysilicon, well regions, metal layers, contacts, vias, and power and ground routing.

The complete physical layout is shown in Figure 3.

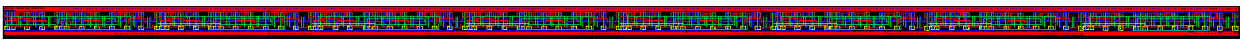
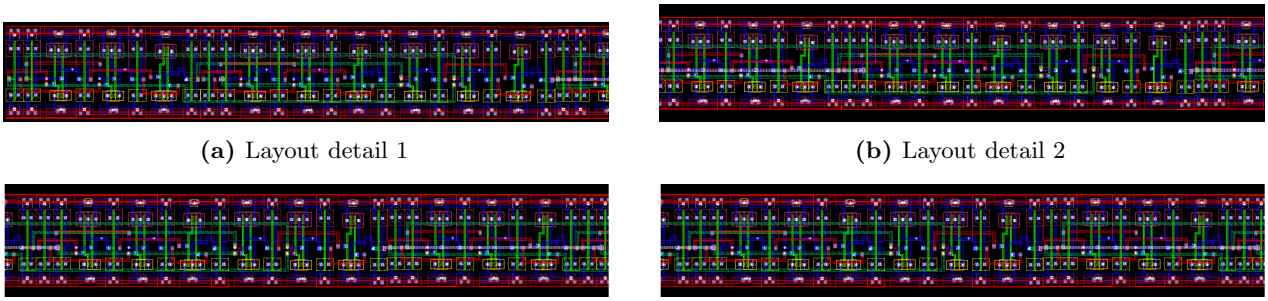


Figure 3: Complete physical layout of the 8-bit ripple carry adder.

6.1 Detailed Layout Views

Figures 4 and 5 present the detailed layout images saved in the project folder. These views provide closer inspection of the physical layout regions and their interconnections.



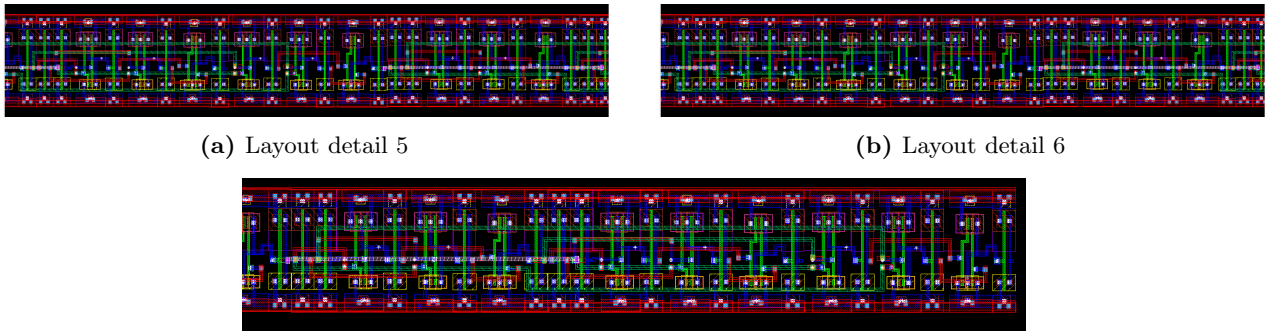
(a) Layout detail 1

(b) Layout detail 2

(c) Layout detail 3

(d) Layout detail 4

Figure 4: Detailed physical layout views (1–4).



(a) Layout detail 5

(b) Layout detail 6

(c) Layout detail 7

Figure 5: Detailed physical layout views (5–7).

7 Design Rule Check (DRC)

Design Rule Check (DRC) is used to determine whether the physical layout satisfies the geometrical fabrication rules of the selected CMOS technology. DRC checks constraints such as minimum metal width and spacing, polysilicon dimensions, diffusion spacing, contact enclosure, via enclosure, and well spacing.

The supplied RCA draft records the DRC result as **PASSED**. Confirm that this matches the final DRC report generated by Cadence before submission.

DRC Result: PASSED

8 Layout Versus Schematic (LVS)

Layout Versus Schematic (LVS) verification compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic. LVS checks circuit connectivity and verifies that the physical layout corresponds to the intended 8-bit ripple carry adder implementation. The verification includes the operand inputs, carry connections between stages, sum outputs, final carry output, power, and ground connections.

The supplied RCA draft records the LVS result as **PASSED**. Confirm that this matches the final LVS report generated by Cadence before submission.

LVS Result: PASSED

9 RCX and Parasitic Extraction

After physical verification, parasitic extraction was performed using the RCX flow. Physical interconnections introduce parasitic resistance and capacitance, which can affect the delay and power consumption of the circuit. The extracted view of the complete 8-bit ripple carry adder is shown in Figure 6.



Figure 6: RCX-extracted view of the 8-bit ripple carry adder.

The extracted view was used for the post-layout transient simulation and performance analysis described in the following sections.

10 Pre-layout Simulation

Pre-layout transient simulation was performed using the schematic view of the 8-bit ripple carry adder in Cadence Virtuoso. This simulation was used to verify the logical behavior of the circuit before considering the parasitic resistance and capacitance associated with the physical layout.

The input sequences were applied to the two 8-bit operands and the initial carry input. The resulting sum and carry outputs were inspected to verify the arithmetic operation. The pre-layout output waveforms and Calculator measurements are presented below.

10.1 Pre-layout Output Waveforms

The pre-layout transient simulation waveforms saved in the project folder are shown in Figures 7–9. The input and output traces are compared with the expected result of the 8-bit addition.

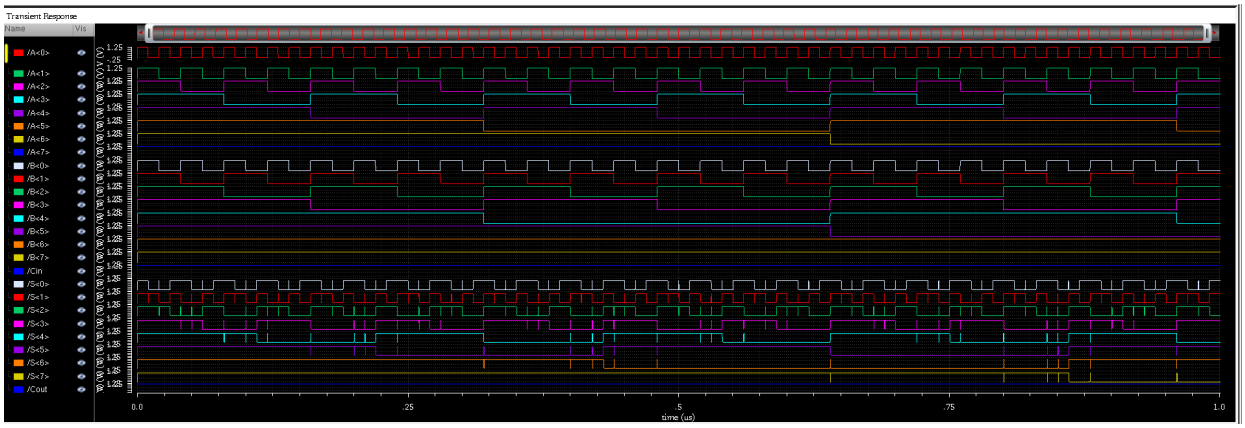


Figure 7: Pre-layout transient waveform view (state 2).



Figure 8: Pre-layout transient waveform set 1 (state 1).

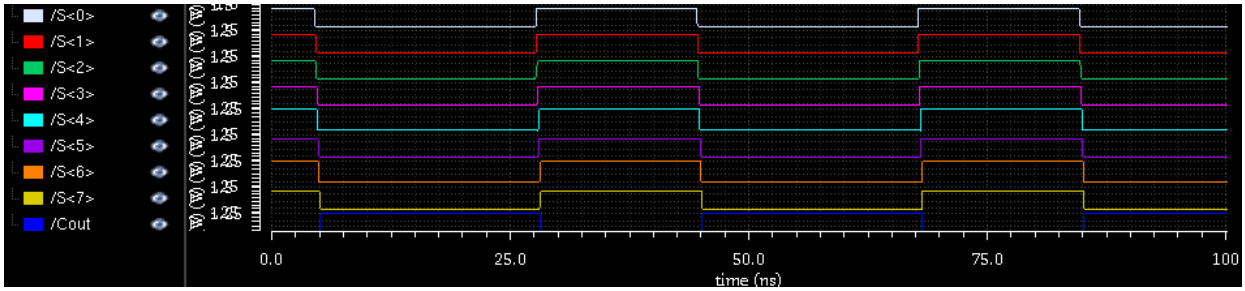


Figure 9: Pre-layout transient waveform set 2 (state 1).

For each applied input combination, the expected arithmetic result is

$$\{C_{out}, S[7:0]\} = A[7:0] + B[7:0] + C_{in}. \tag{11}$$

The waveform traces are checked against this relation to verify the functional operation of the adder.

10.2 Pre-layout Average Power Calculation

The pre-layout average power consumption was obtained from the Cadence Virtuoso Calculator using the schematic-level transient simulation. The average power over a measurement interval from t_1 to t_2 is expressed as

$$P_{avg,pre} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} p_{pre}(t)\, dt, \tag{12}$$

where $p_{pre}(t)$ represents the instantaneous power measured in the pre-layout simulation. The corresponding Calculator measurement is shown in Figure 10.

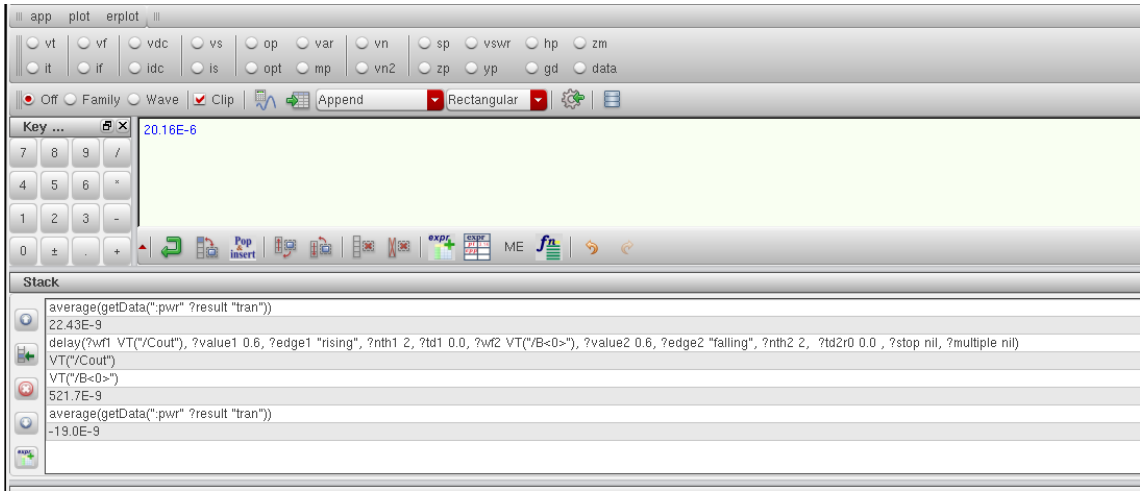


Figure 10: Pre-layout average power measurement using the Cadence Virtuoso Calculator.

The measured pre-layout average power consumption is

$$P_{avg,pre} = 20.12 \times 10^{-6}\,\mathrm{W} = 20.12\,\mu\mathrm{W}. \tag{13}$$

Therefore, the measured pre-layout average power consumption of the 8-bit ripple carry adder is approximately $20.12\,\mu$W under the applied simulation conditions.

10.3 Pre-layout Worst-case Propagation Delay

Propagation delay represents the time required for an input transition to produce the corresponding output transition. For the ripple carry adder, the measured worst-case delay was recorded from the pre-layout transient simulation using the Cadence Virtuoso Calculator.

For a selected input-to-output transition, the propagation delay is

$$t_p = t_{out} - t_{in}, \tag{14}$$

where t_{in} and t_{out} are the times at which the input and output cross the selected measurement threshold. The pre-layout worst-case delay measurement is shown in Figure 11.

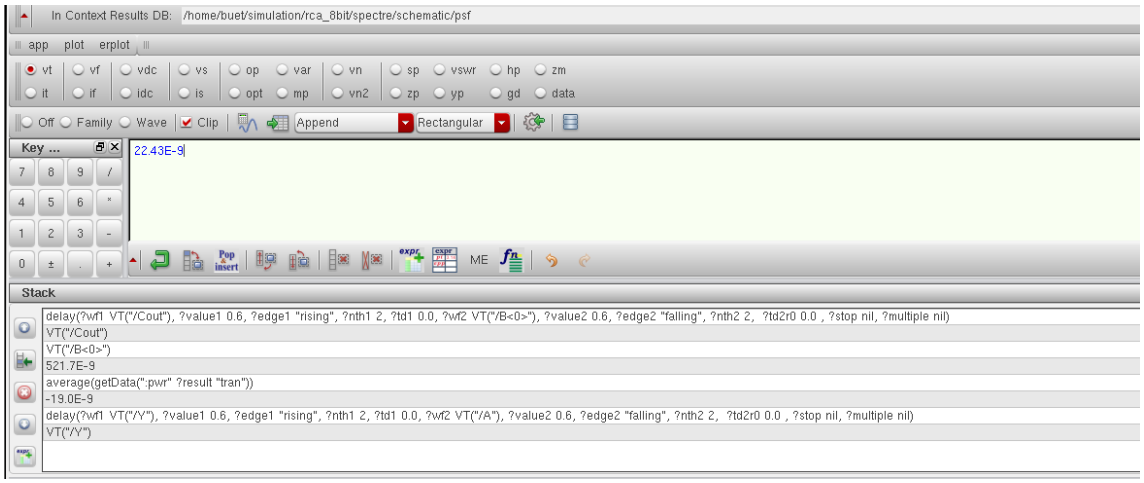


Figure 11: Pre-layout worst-case propagation-delay measurement using the Cadence Virtuoso Calculator.

The measured pre-layout worst-case propagation delay is

$$t_{pd,pre} = 22.43 \times 10^{-9}\,\mathrm{s} = 22.43\,\mathrm{ns}. \tag{15}$$

11 Post-layout Simulation

Post-layout transient simulation was performed using the RCX-extracted view of the 8-bit ripple carry adder in Cadence Virtuoso. Unlike the pre-layout simulation, the extracted circuit includes the parasitic resistance and capacitance associated with the physical interconnections.

The post-layout simulation was used to verify the operation of the extracted circuit and to evaluate its average power consumption and worst-case propagation delay. The output waveforms and Calculator measurements are presented in the following sections.

11.1 Post-layout Output Waveforms

The post-layout transient waveforms of the extracted 8-bit ripple carry adder are shown in Figures 12 and 13. These waveforms are used to examine the sum and final carry outputs while accounting for the extracted interconnect parasitics.

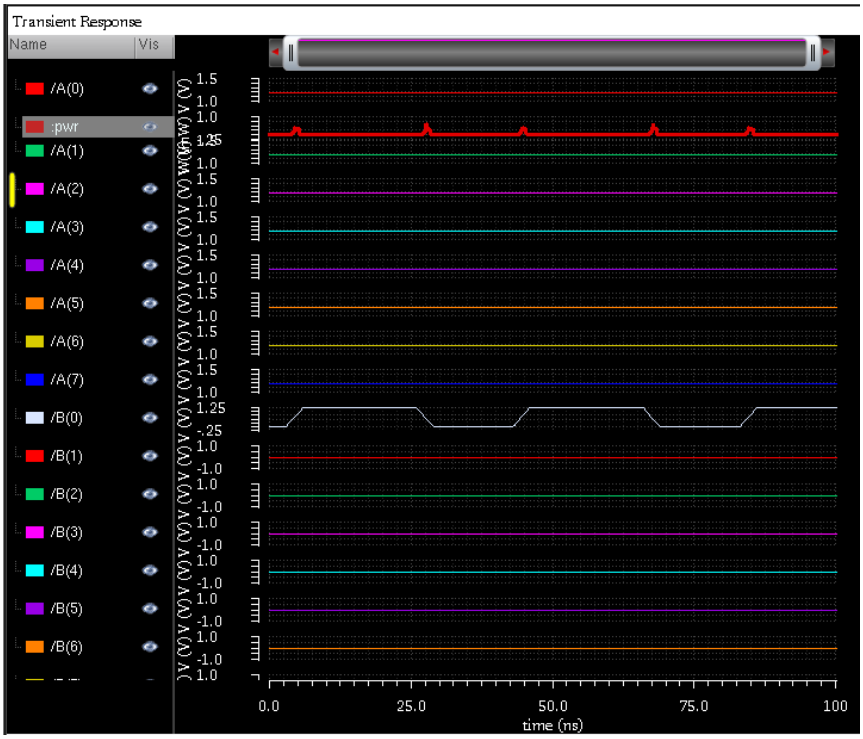


Figure 12: Post-layout transient waveform set 1 (state 3).

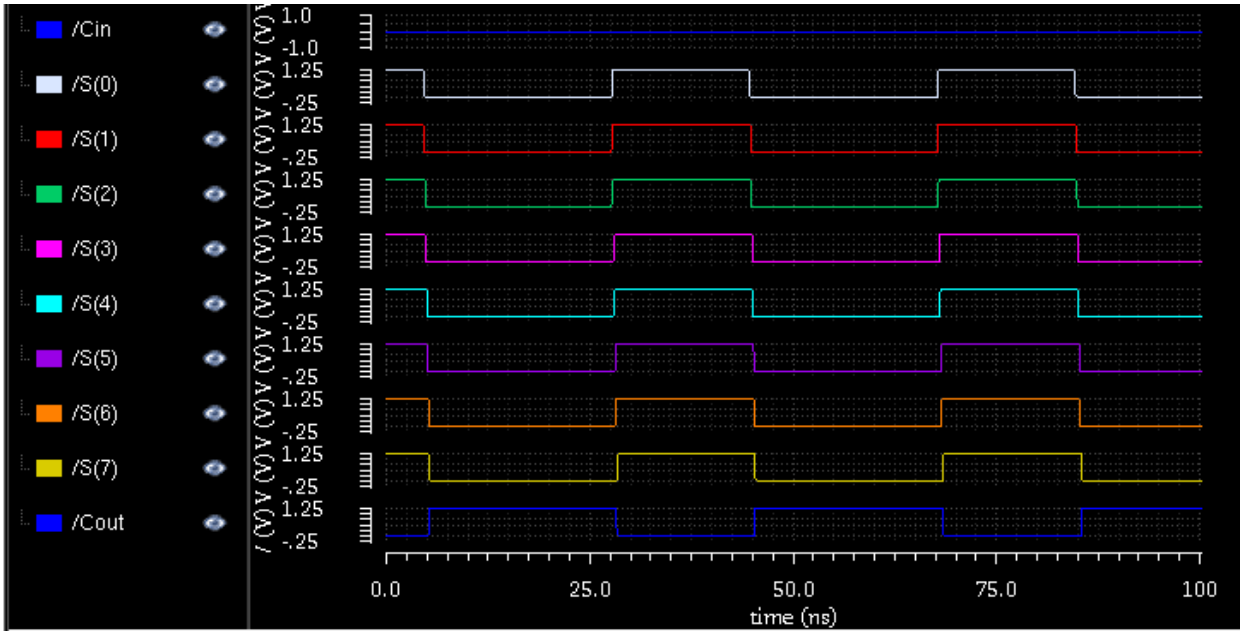


Figure 13: Post-layout transient waveform set 2 (state 3).

The output is expected to satisfy

$$\{C_{out}, S[7:0]\} = A[7:0] + B[7:0] + C_{in}. \tag{16}$$

The simulated traces are compared with the expected arithmetic result to check whether the extracted circuit maintains the intended functionality.

11.2 Post-layout Average Power Calculation

The post-layout average power consumption was calculated from the transient simulation of the RCX-extracted circuit using the Cadence Virtuoso Calculator. The average power over the selected measurement interval is

$$P_{avg,post} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} p_{post}(t)\, dt, \tag{17}$$

where $p_{post}(t)$ represents the instantaneous power consumption of the extracted circuit. The post-layout power calculation is shown in Figure 14.

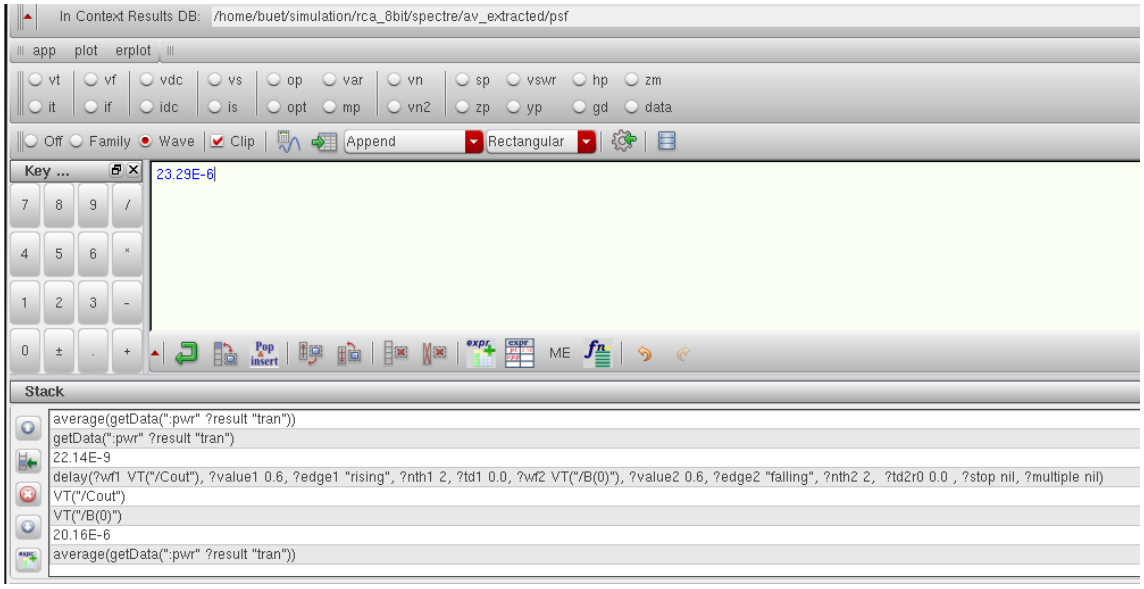


Figure 14: Post-layout average power measurement of the RCX-extracted 8-bit ripple carry adder.

The measured post-layout average power consumption is

$$P_{avg,post} = 23.29 \times 10^{-6}\,\mathrm{W} = 23.29\,\mu\mathrm{W}. \tag{18}$$

Therefore, the measured post-layout average power consumption is approximately $23.29\,\mu$W under the applied simulation conditions.

11.3 Post-layout Worst-case Propagation Delay

The post-layout worst-case propagation delay was measured using the transient response of the RCX-extracted circuit. The extracted parasitic resistance and capacitance can affect the transition time of the circuit signals. For the selected input-to-output transition, the propagation delay is calculated as

$$t_{pd,post} = t_{out} - t_{in}. \tag{19}$$

The delay measurement is shown in Figure 15.

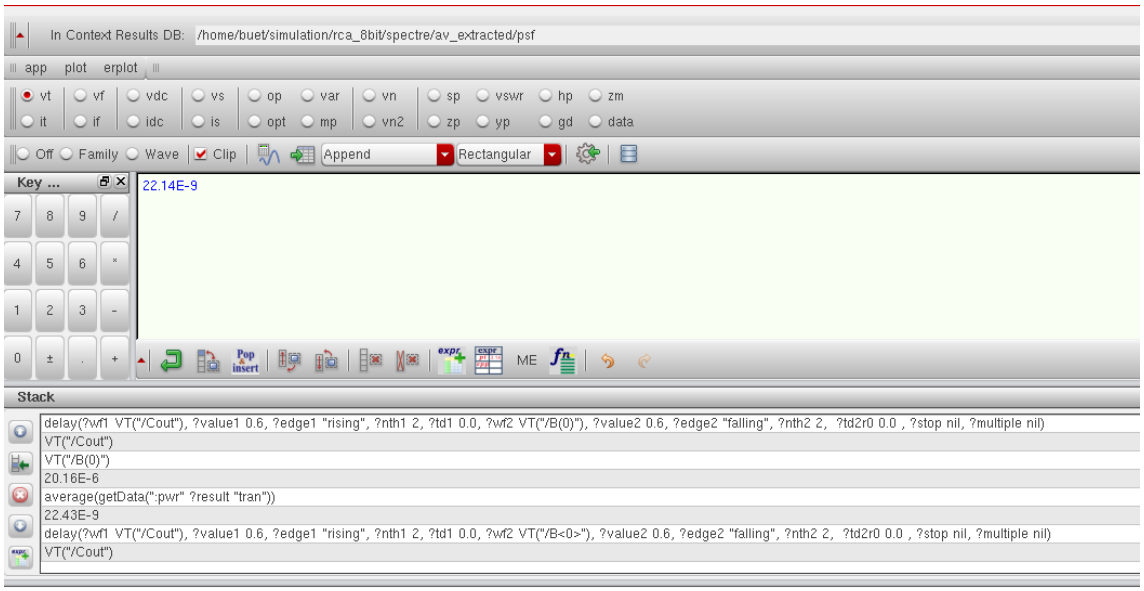


Figure 15: Post-layout worst-case propagation-delay measurement of the extracted 8-bit ripple carry adder.

The measured post-layout worst-case propagation delay is

$$t_{pd,post} = 22.14 \times 10^{-9}\,\mathrm{s} = 22.14\,\mathrm{ns}. \tag{20}$$

12 Pre-layout and Post-layout Performance Comparison

The pre-layout and post-layout results are compared to examine the change in measured performance after physical layout and parasitic extraction.

The percentage change in average power is calculated as

$$\%\Delta P = \frac{P_{avg,post} - P_{avg,pre}}{P_{avg,pre}} \times 100\%. \tag{21}$$

Substituting the measured values,

$$\Delta P = 23.29 - 20.12 = 3.17\,\mu\mathrm{W}, \tag{22}$$

$$\%\Delta P = \frac{3.17}{20.12} \times 100\% \approx 15.76\%. \tag{23}$$

Thus, the post-layout average power increased by $3.17\,\mu$W, or approximately 15.76%, compared with the pre-layout result.

Similarly, the percentage change in worst-case propagation delay is

$$\%\Delta t_{pd} = \frac{t_{pd,post} - t_{pd,pre}}{t_{pd,pre}} \times 100\%. \tag{24}$$

Substituting the measured values,

$$\Delta t_{pd} = 22.14 - 22.43 = -0.29\,\mathrm{ns}, \tag{25}$$

$$\%\Delta t_{pd} = \frac{-0.29}{22.43} \times 100\% \approx -1.29\%. \tag{26}$$

The negative sign indicates that the measured post-layout worst-case delay is lower. Therefore, the recorded delay decreased by 0.29 ns, corresponding to an improvement of approximately 1.29%; it did not increase.

Table 4: Comparison of pre-layout and post-layout performance

Parameter	Pre-layout	Post-layout	Change
Average power	$20.12\,\mu$W	$23.29\,\mu$W	$+3.17\,\mu$W (+15.76%)
Worst-case propagation delay	22.43 ns	22.14 ns	−0.29 ns (−1.29%)

The comparison is meaningful when the pre-layout and post-layout simulations use the same input pattern, measurement interval, supply conditions, and delay definition. The slightly lower measured post-layout delay should therefore be interpreted with reference to the exact input-to-output transition, threshold, and Calculator expressions used for both measurements.

13 Verification and Performance Summary

The main design, verification, and simulation results of the 8-bit ripple carry adder are summarized in Table 5.

Table 5: Summary of the 8-bit ripple carry adder design results

Parameter	Result
Architecture	8-bit ripple carry adder
Full-adder stages	8 cascaded stages
Operand inputs	$A[7:0]$, $B[7:0]$
Carry input	C_{in}
Outputs	$S[7:0]$, C_{out}
Schematic design	Completed
Symbol generation	Completed
Physical layout	Completed
DRC	Passed (as recorded in the supplied draft; verify final report)
LVS	Passed (as recorded in the supplied draft; verify final report)
RCX / parasitic extraction	Extracted view generated
Pre-layout simulation	Completed
Post-layout simulation	Completed
Pre-layout average power	$20.12\,\mu$W
Post-layout average power	$23.29\,\mu$W
Pre-layout worst-case delay	22.43 ns
Post-layout worst-case delay	22.14 ns

14 Discussion

An 8-bit ripple carry adder was designed using eight cascaded 1-bit full-adder stages in Cadence Virtuoso. Each stage adds the corresponding operand bits and the carry received from the previous stage. The sum and carry outputs are described by

$$S_i = A_i \oplus B_i \oplus C_i, \tag{27}$$

$$C_{i+1} = A_i B_i + C_i(A_i \oplus B_i). \tag{28}$$

The schematic and generated symbol represent the intended 8-bit arithmetic circuit. The physical layout was developed, and the DRC and LVS results are recorded in the respective sections. The RCX-extracted view was used for post-layout transient simulation.

The pre-layout average power was measured as $20.12\,\mu$W, while the post-layout average power was $23.29\,\mu$W. Thus, the average power increased by $3.17\,\mu$W, or approximately 15.76%, after layout extraction.

The measured worst-case propagation delay changed from 22.43 ns before layout extraction to 22.14 ns after extraction. This represents a decrease of 0.29 ns, or approximately 1.29%. The recorded post-layout result therefore shows a small delay reduction rather than an increase. This difference should be interpreted with reference to the applied input patterns, measurement interval, supply conditions, selected input-to-output transition, threshold, and Calculator expressions used for both measurements.

The ripple carry architecture has a straightforward and modular structure. However, its carry signal propagates sequentially from one stage to the next. Therefore, the carry-chain path is an important factor in the worst-case delay of the 8-bit adder.

15 Conclusion

An 8-bit ripple carry adder was designed and analyzed using Cadence Virtuoso. The circuit consists of eight cascaded full-adder stages and produces an 8-bit sum and a final carry output from two 8-bit operands and an initial carry input.

The project covered schematic design, symbol generation, physical layout design, DRC, LVS, RC parasitic extraction, and pre-layout and post-layout transient simulation. The output waveforms were examined to verify the expected arithmetic behavior.

The measured pre-layout average power consumption was $20.12\,\mu$W, while the post-layout average power consumption was $23.29\,\mu$W. The measured pre-layout and post-layout worst-case propagation delays were 22.43 ns and 22.14 ns, respectively. The post-layout power increased by approximately 15.76%, whereas the measured worst-case delay decreased by approximately 1.29%.

Overall, this project demonstrates the design, physical implementation, verification, parasitic extraction, and performance analysis of an 8-bit ripple carry adder using Cadence Virtuoso. The final DRC and LVS statuses should be consistent with the actual Cadence verification reports.

16 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator